WEEK -10

Minor Project (BTCS607N)-Weekly Progress Report

Project Title: Drone-Based Intelligent System for Apple Orchard Management in Himachal Pradesh.

Group ID: G9

Branch & Section: BTech CSE (AIML with Microsoft), AIDS -A

Date: 10-04-2026

1. Executive Summary

A. **Current Sprint:** Sprint 1 (Issue – 10)

B. **Sprint / Issue Goal:** To develop frontend UI, integrate it with backend APIs, and connect the application with the database for complete data flow.

C. **Status:** On Track

2. UML & OOAD Progress

A. **Diagrams Completed this week:** No new diagrams created

B. **Diagrams Revised:** Updated system architecture based on implementation needs

C. **OO Concept Applied:**

1. Layered Architecture (Frontend, Backend, Database separation)
2. Modular Design (routes, services, utils separation)
3. MVC Pattern (UI → API → Database)
4. Service-based architecture for prediction and advisory modules

3.Technical Execution (DevOps & Git)

A. **Jira Tasks Completed:**

1. Developed frontend pages (login, signup, dashboard, upload, result)
2. Connected UI with backend APIs
3. Integrated database for data storage and retrieval
4. Implemented authentication and validation
5. Tested API endpoints
6. Fixed database schema and integration issues

Link://ayushshrivastava.atlassian.net/jira/software/projects/DBISFAOM/boards/2?atlOrigin=eyJpIjoiY2I0Y2FjMmQ2MTliNDBlM2FjYzEyN2U1MjdkY2QzNGMiLCJwIjoiaiJ9

B. **Git Activity:**

1. Added frontend files (HTML pages)
2. Updated backend integration code
3. Committed and pushed changes to repository

Link : https://github.com/yp1818/Drone-Based-Apple-Orchard-Management

C. **CI/CD Status:**

1. Implemented basic CI using GitHub Actions for build/testing
2. Automated deployment pipeline (if applicable)

4. Updated System Architecture

```
a:\Drone Apple Orchard Management System
|
+--- README.md
|
+--- Backend\
|   |--- database.py
|   |--- main.py
|   |--- models.py
|   |--- requirements.txt
|   |--- schemas.py
|   |
|   +--- model\
|   |      best.pt
|   |
|   +--- routes\
|   |      auth_route.py
|   |      predict_route.py
|   |      feedback_route.py
|   |      predict_route.py
|   |
|   +--- services\
|   |      advisory_service.py
|   |      prediction_service.py
|   |
```

```
|   +--- static\
|   |   \--- uploads\
|   |          blackrot_080.jpg
|   |          rust_007.jpg
|   |          (and 16 other uploaded sample JPGs)
|   |
|   +--- test\
|   |      test_model.py
|   |      test_api.py
|   |      test_local.py
|   |      test_verify.py
|   |
|   \--- utils\
|   |       file_handler.py
|   |       logic.py
|   |
+--- frontend\
|   |--- dashboard.html
|   |--- index.html
|   |--- login.html
|   |--- result.html
|   |--- signup.html
|   |--- upload.html
|   |
|   +--- assets\
|   |   +--- icons\
|   |   \--- images\
|   |          orchard-bg.webp
|   |
|   +--- css\
|   |      style.css
|   |
|   \--- js\
|          app.js
```

5.Testing Summary

1. Tested API endpoints using Postman
2. Verified frontend to backend communication
3. Checked data storage and retrieval from database
4. Debugged integration issues between UI, backend, and database
5. Ensured proper authentication and input validation

6.Goals for Next Week

1. Perform Unit Testing using PyTest to verify individual functions and modules
2. Test backend methods to ensure they work as expected
3. Conduct Integration Testing to check the flow between:
    - Frontend → Backend
    - Backend → Database
    - Backend → ML model
4. Validate complete system workflow (end-to-end testing)
5. Debug and fix issues found during testing
6. Generate and analyze test reports

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